

A COMPARATIVE ANALYSIS BETWEEN SIGNATURE-BASED TECHNIQUE AND BEHAVIOR-BASED TECHNIQUE IN MALWARE DETECTION

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Introduction

Since the internet has been playing a crucial role in the modern day, especially after the COVID-19 outbreak, more businesses have been shifting to work in remote or hybrid.

Coming with the development of the internet; malware attack has become a great concern for businesses as it has been rising dramatically, malware is used by hackers to steal data, gain unauthorized access or/and control. To prevent the risk from malware, malware detection system is being used to scan, analyze and identify possible malware in software, network traffic. This research will compare the advantages and disadvantages between signature-based malware detection technique and behavior-based malware detection technique in terms of speed, accuracy rate.

Signature-based detection approach

Signature-based technique is widely used in commercial anti-virus software, it marks a program benign or malicious by comparing the program's signature with the signature database, the signature database contains the signatures of known malwares (P. Ioulianos et al., 2018, para. 1). The signature of the program could be a sequence of bytes in the code, API calls, opcodes, and entropy quantity (A. Aboaoja et al. 2022, p. 10; P. et al., n.d., p. 75; Tahir, 2018, p. 26). Signature-based technique detects malware by scanning each file of the program, in every file, there are signatures generated, these signatures will be compared with the known signatures in the database. Sharma and Singh (2024) have explained 7 phases in the working process of signature-based malware detection, in the first stage, the researchers examine known malware and extract their unique traits such as behavioral patterns, file structures, code snippets, those are called

signatures, these signatures will then be stored in a centralized database. Next phase, when the user starts a malware scan, the anti-virus software will scan every application, file in the user's device, this is the scanning process. In the fourth phase, all files and applications' signature will be compared with the signed found by the researchers stored in the database. The next phase will detect if the signature scanned is match with known malwares' signatures. The sixth phase will only happen if the application or file is marked as malware, depending on the user's configuration, the anti-virus software will perform a suitable action to stop the malicious

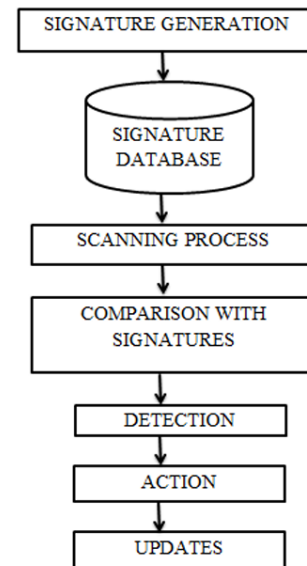


Figure 1: The process of signature-based malware detection. From Sharma & Singh, 2024, p. 3

program from running. The last phase is keeping the signature database up to date (p. 3). There are also multiple techniques to generate a signature such as string scanning, top-and-tail scanning, entry-point scanning, integrity checking (Aslan & Samet, 2020, pp. 6254-6255).

Behavior-based detection approach

On the other hand, behaviour-based detection technique observes the behavior of the program before marking if the program is malicious or benign, anomaly detection technique is a major technique under behavior-based malware detection technique (Chakravarty et al. n.d., p.1510; A. Saeed et al., 2013, p.28). The system will monitor the program's behavior, normal behavior is stored as reference, if the program behavior is not matched with the reference, the program is marked as malicious (Chakravarty et al. n.d., p.1510; A. Saeed et al., 2013, p.28). The behavior observed during the execution are the system calls that the program makes with the OS. (Chakravarty et al.

n.d., p.1510). Malicious behavior is discovered using dynamic analysis to evaluate malicious intent based on the program's code and structure. Data mining algorithms in machine learning can be applied on behavior-based approach, it uses feature extraction to extract API calls and Assembly features to train the machine learning model (Souri & Hosseini, 2018, p.5).

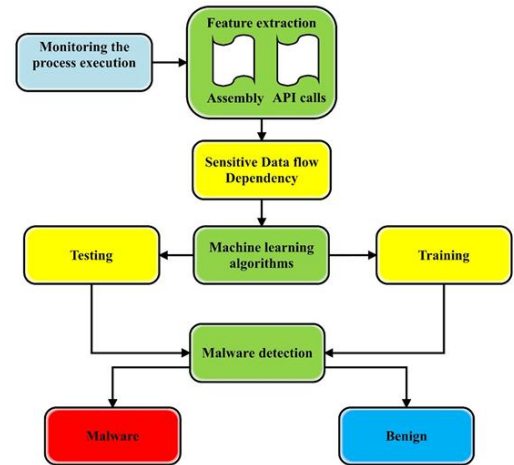


Figure 2: The behavior-based detection framework. From Souri & Hosseini, 2018, p.5

Comparison

The difference between signature-based technique and behavior-based technique is signature-based technique uses classification to determine if a program is a malware, while behaviour-based uses pattern in the program's activities to determine. Each technique has its own strength and drawbacks.

Strength and drawbacks of signature-based detection approach

According to Kumar & Sangwan (2012), this detection technique offers low false-positive rate if the attack is clearly defined and the technique is also easy to use and setup. However, this technique also has disadvantages. This approach is likely failed to detect zero-day attack, the technique also raise alert regardless of the outcome, for example, if a failed attack is performed on the system, the detector still raises alert about the attack no matter of if the attack was successful or failed. This approach will also not be practical for inside attack that involve abuse of privileges (p.35).

Suri and Hosseini (2018) noted that signature-based detection offers lower execution time, broadly accessible and it also decreases system overhead. However, the main disadvantage of this

approach is it omits feature selection, besides signature-based detection requires signature for each malware variation which makes the database grows at an exponential rate (pp.3-4; p.8). According to Li et al. (2017) and Anuradha (2019) feature selection is to select the subset of relevant features and to clean up the noise and irrelevant data (p.2; p.3), in this context, feature selection helps identify the pattern of malware, and signature-based technique skips this step, so that malicious program with obfuscation techniques such as polymorphism, which encrypts the binary code of the malware might pass the signature check (O. Or-mel et al., 2019, p.16; Ye et al., 2017, p.7).

Idika & Mathur (2007) argued because of the dependence on human expertise, this technique also might have human errors, and it could take time for the expertise to analyze the malware before adding its signature to the database, this is considerably amount of time compared to the speed of spreading of a malware, they also argued that signature-based detection technique is vulnerable to zero-day attack. (p.32).

Strength and drawbacks of behavior-based detection approach

According to Sourì and Hosseini (2018), the main advantage of behavior-based detection is the approach can detect zero-day malwares, as well as polymorphic malwares. However, the disadvantages of this approach are time complexity and storage complexity for behavioural patterns (p.5; p.13)

A. Aboaoja et al. (2022) pointed out some disadvantages of behavior-based malware detection that the approach can misclassifies the behaviour between malware and benign software which leads to the high rate of false-positive. Moreover, malware is able to recognize the nature of its executing environment using evasion techniques, hence, it can change the behaviors to be benign-like behavior or terminate its execution to bypass the detection approach. Furthermore, some

researchers trained their models with recent malwares, as a result some models become vulnerable with older malicious behaviours (p.14)

Comparison table

	Signature-based	Behavior-based
Advantages	- Simplicity - Low false-positive rate - Low execution time - Lower system overhead	- Ability to detect zero-day attack and malicious program with obfuscation techniques
Disadvantages	- Fail to detect zero-day attack - Not practical for inside attacks - Database growth - Malicious program with obfuscation techniques might pass the signature check - Depends on human experts	- High false-positive rate - Complexity - Malicious program can pass the behavior check if it can change its behavior to benign-like or terminate its execution - Some models is less likely to be able to detect older malicious behaviors

Conclusion

In the modern world when business operation is not only happening inside the office but also on the internet, choosing the right tools to protect against threats can help organization to minimize risks and to save money and resources. This research paper has analyzed the benefits and drawbacks of 2 malware detection methods. No tool has brought the perfect solution, while signature-based approach offers simplicities, precise detection with known threats, behavior-based approach offers the adaptability to the business with unknown threats. A combination between two tools could be developed to take the simplicity and precise of signature-based approach, and the adaptability and flexibility with new threats of behavior-based approach.

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